

Problem 1. Translate these statements into logic using the provided propositions.

$v \equiv$ The message is scanned for viruses.

$s \equiv$ The message is marked as spam.

$k \equiv$ The message was sent from a known system.

$b \equiv$ The sender is blocked.

- a) The message is scanned for viruses or marked as spam.
- b) A message from a blocked sender goes scanned for viruses.
- c) The sender is blocked and the message is scanned for viruses only if the message was sent from an unknown system.
- d) If the message is sent from an unknown system, either the sender is blocked or the message is marked for spam. In both cases, it is scanned for viruses.

Answer.

- a) $v \vee s$
- b) $b \rightarrow v$
- c) $(b \wedge v) \rightarrow \neg k$
- d) $\neg k \rightarrow ((b \vee s) \wedge v)$

Problem 2. Prove the following by using **logical equivalences** (i.e., using direct manipulation). For each step, state which single logical equivalence was used.

- a) $\neg(p \rightarrow \neg q) \equiv (p \wedge q)$
- b) $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Answer.

- a) $\neg(p \rightarrow \neg q) \equiv (p \wedge q)$
 $\neg(\neg p \vee \neg q) \equiv p \wedge q$ implication equivalence
 $p \wedge q \equiv p \wedge q$ demorgan's law
- b) $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$
 $p \rightarrow (\neg q \vee r) \equiv (p \wedge q) \rightarrow r$ implication equivalence
 $\neg p \vee (\neg q \vee r) \equiv (p \wedge q) \rightarrow r$ same as above
 $\neg p \vee \neg q \vee r \equiv (p \wedge q) \rightarrow r$ ungroup
 $\neg(p \wedge q) \vee r \equiv (p \wedge q) \rightarrow r$ demorgan's law
 $(p \wedge q) \rightarrow r \equiv (p \wedge q) \rightarrow r$ implication equivalence

Problem 3. Use the truth tables to:

- a) Show that $p \vee (p \wedge q) \equiv p$
- b) Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent

Answer.

a)

p, q	$p \wedge q$	$p \vee (p \wedge q)$	p
$\perp \perp$	$\perp$	$\perp$	$\perp$
$\perp \top$	$\perp$	$\perp$	$\perp$
$\top \perp$	$\perp$	$\top$	$\top$
$\top \top$	$\top$	$\top$	$\top$

b)

p, q, r	$p \wedge q$	$(p \wedge q) \rightarrow r$	$p \rightarrow r$	$q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
$\perp \perp \perp$	$\perp$	$\top$	$\top$	$\top$	$\top$
$\perp \perp \top$	$\perp$	$\top$	$\top$	$\top$	$\top$
$\perp \top \perp$	$\perp$	$\top$	$\top$	$\perp$	$\perp$
$\perp \top \top$	$\perp$	$\top$	$\top$	$\top$	$\top$
$\top \perp \perp$	$\perp$	$\top$	$\perp$	$\top$	$\perp$
$\top \perp \top$	$\perp$	$\top$	$\top$	$\top$	$\top$
$\top \top \perp$	$\top$	$\perp$	$\perp$	$\perp$	$\perp$
$\top \top \top$	$\top$	$\top$	$\top$	$\top$	$\top$

Problem 4. Let $M(x) \equiv x \text{ visits the Carnegie Museum}$ and $A(x) \equiv x \text{ is an art student}$. Let the domain of each be “all students at Pitt”.

- a) Translate the following statement into English.

$$\forall x [M(x) \rightarrow A(x)]$$

- b) translate the following into predicate logic.
“not all art students have visited the carnegie museum”.
- c) Explain why the statement “Some art students visit the Carnegie Museum” cannot be translated as:

$$\exists x [A(x) \rightarrow M(x)]$$

Answer.

- a) For all students at pitt, if they visit the Carnegie Museum, they are an art student.
- b) $\neg(\forall x [A(x) \rightarrow M(x)])$, which is equivalent to $\exists x [A(x) \wedge \neg M(x)]$
- c) The proposition given indicates the relationship between art student and whether they visit the Carnegie Museum. It states that if they are an art student, they will visit the museum. However, that is not the same as the statement given. The correct proposition should be: $\exists x [A(x) \wedge M(x)]$, where it correctly represents the status of being an art student and if they visited the museum.

Problem 5. Explain the following logical paradoxes. That is, explain why each one cannot be assigned a specific value.

- a) “The next statement is true. The previous statement is false.”
- b) “If this sentence is true, then all statements are true.”
- c) Define two propositional variables as $p \equiv q$ is false, and $q \equiv p$ is false.

Answer.

- a) These two statements contradict each other. We can first rule out the case that they are both true or false, since that would violate the statements. For the first case, we assume the first statement is true; if it is true, then according to the statement, the second statement will be true; however, if the second statement is true, then the first statement is false, which contradicts our assumption. For the second case, we assume the second statement is true. If the second statement is true, then the first statement is false. However, if the first statement is false, then the second statement is also false, which also contradicts our assumption. Therefore, we cannot assign a truth value to both propositions.
- b) We can consider two cases: the first part of the proposition is true or false. In the first case, we consider that it is true, which implies the following proposition also have to be true in order for the whole proposition to stay true (since $\top \rightarrow \top = \top$). However, that is not the case: for example, the statement that “I am currently on the planet saturn” is a false statement. Since the second part of the proposition is false, that means we have a true value implicating a false value ($\top \rightarrow \perp = \perp$), which makes this proposition contradict our assumption (as well as itself since it states that all statements are true). For our second case where we consider the first part of the proposition as false, in order for the whole proposition to stay true, the second part of the proposition can either be true or false. Let’s first consider where it is true – however this has already been proven false earlier(“planet saturn”). If the statement is false, however, it means that all statements are false, which is also false. I can make a statement like “The date September 10 in 2025 is a Wednesday” and it is true. Since we can’t assign a truth value to the second statement that “all statements are true”, we also could not assign a truth value to the entire implication.
- c) Let’s assume p is true. If p is true, then q is false. If q is false, then it implies that p is true. This case stands. On the other hand, if p is false, then q is true; and if q is true, then p is false. This case also stands. Since both case stands, we cannot come to a conclusion of which truth value can be assigned to p or q .